

Tuesday 9AM - 12PM

The effect of varying amplitude and frequency of vertical input vibration on the stability of an inverted pendulum system

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2.671 Measurement and Instrumentation

Tuesday AM

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Abstract

Actuating an inverted pendulum rapidly up and down can, rather unexpectedly, make it stays upright. An inverted pendulum setup with vertical excitation is created to investigate the behavior of such system. A speaker is used to vibrate a 6.5 cm plastic rod to obtain acceleration results across the range of frequency between 16 and 30 Hz. The trials are repeated 5 times for each frequency, and the acceleration data is then processed to find the position measurement. The obtained experimental data is used to compare with a current theory presented by Landau and Lifshitz. The constant of fit for both stability condition with 95% confidence interval is $a_{local} = 0.1420 \pm 0.0045 [m Hz]$ and $a_{global} = 0.1491 \pm 0.0060 [m Hz]$, which has a difference of 3% and 2% with the empirical model, respectively. Thus, the theory provides a good prediction to the stability behavior of the inverted pendulum system.

Keywords

Inverted pendulum, Dynamical stabilization, Kapitza's pendulum, Vibration

1. Introduction

Almost everyone has tried to balance a stick on the tip of their hand. The stick is unstable, and requires hand movements to counteract the falling motion of the pendulum. This is known as inverted pendulum system, which is ubiquitous and familiar mechanical system. Yet, behind the simplicity of such setup encloses many fascinating behaviors that are hidden in plain sight. One such instance can be seen when subjecting the rod to rapid oscillation. If the rod is jigged vertically, it can also stay upright by itself given sufficient condition (see Figure 1). Even more impressively, a system of multiple pendulums can also be stabilized this way. These systems are known in literature as the “Kapitza's Pendulum” [1][2][3][4].



Figure 1: Illustration of an inverted pendulum system stabilized by a sabre saw. The pendulum stays upright due to the rapid oscillation being applied at the pivot (provided by the saw). The experiment tests one of the theories that explains the pendulum behavior. Image from [1]

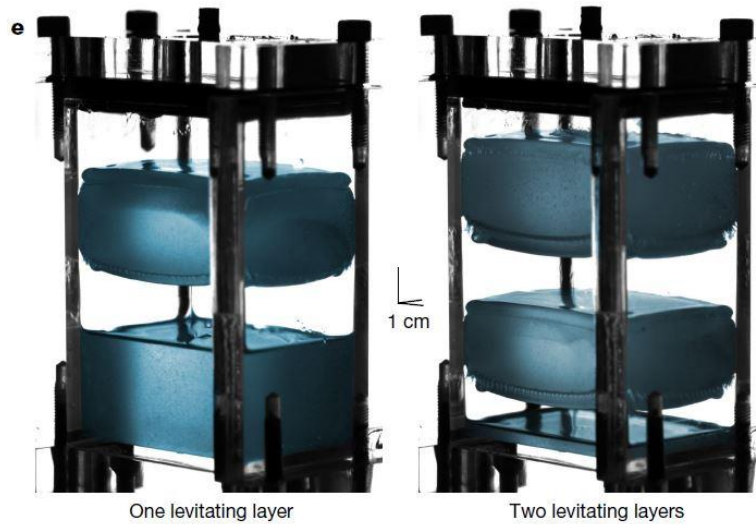


Figure 2: A viscous fluid being levitated by applying vertical vibration to the container. The principle behind Kapitza's pendulum can be applied to explain dynamical stability of various physical systems. Image from [5].

This behavior can be extended to give insights into other fascinating dynamical behaviors. One such example is the fluidic experiment being done by a group of researchers at *Université de Paris* and *Institut Langevin* in 2020, where the effect of vibration on viscous fluid is studied. The study shows a column of a viscous liquid being *levitated* in the air by vibrating the container rapidly (the demonstration can be viewed online at the author's website) [4][5]. The liquid is shown to be suspended in the air and floating (See Figure 2). Understanding the behavior of the Kapitza's pendulum setup is a good step toward understanding these dynamical systems.

To explore the behavior of such system, deriving equation of motion that describes the motion of the pendulum is essential for further analysis. The stability behavior of a Kapitza's pendulum can then be explored from this equation of motion, and there are several methods of analyzing such system. One simplified model is described in Landau and Lifshitz's *Course of Theoretical Physics* book, where the method of defining a small oscillatory motion can be used to derive the stability condition for system undergoes rapid oscillation [8]. This model is chosen to compare with the experiment due to its simplicity, as well as the ease of verification with experimental data. *For a complete treatment, more advanced techniques for analyzing stability of differential equation such as Mathieu's solution, Hill determinant and Floquet equation can be referenced* [2][3].

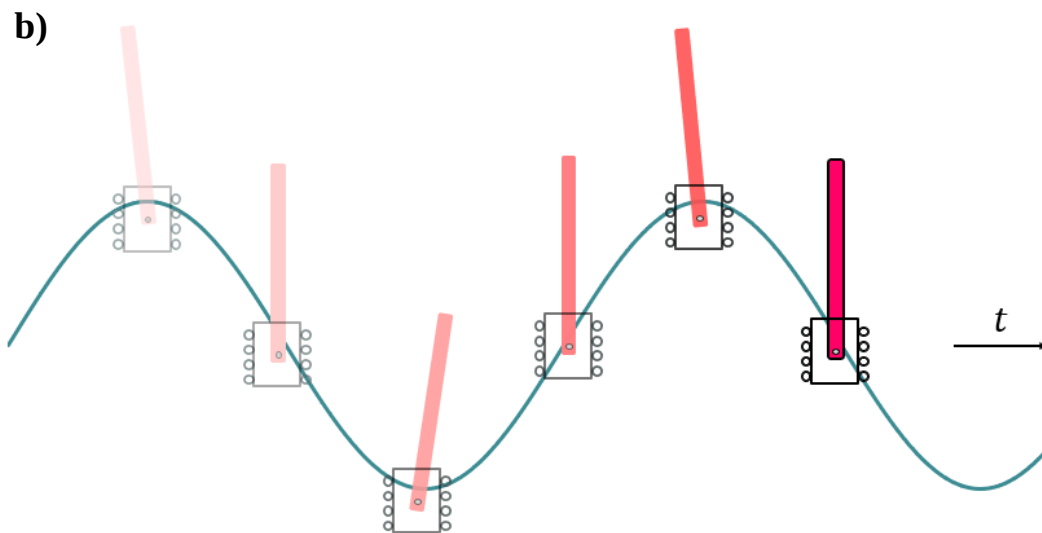
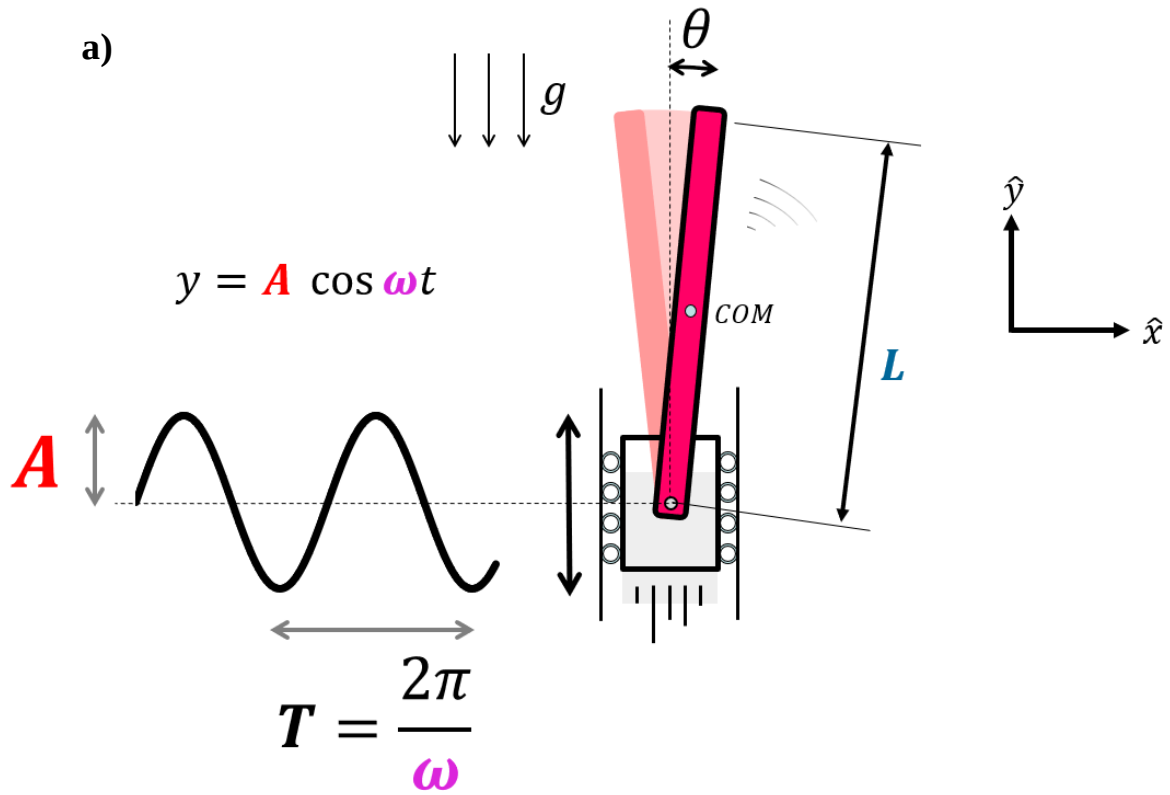


Figure 3: Illustration of a Kapitza's pendulum setup. Figure 3a): An inverted pendulum with vibrating pivot can maintain its stability. Figure 3b) Time lapses of pendulum movement.

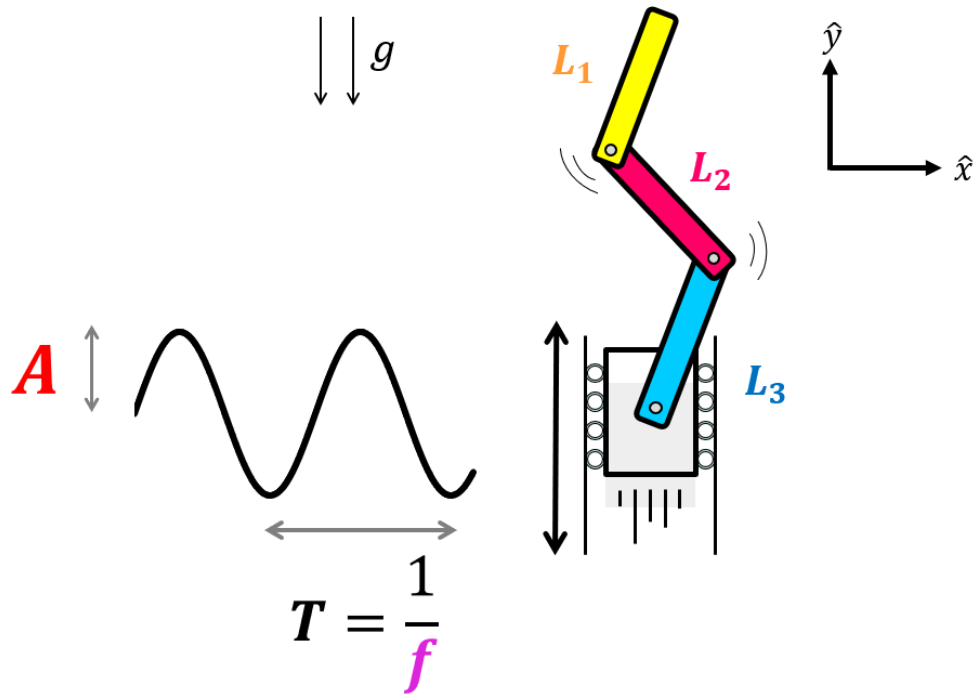


Figure 4: Illustration for a multiple pendulum setup. Such system can also be stabilized if sufficient vibration is provided.

In order to compare with the chosen mathematical model for a single pendulum, the input frequency and amplitude on the stability of an inverted pendulum with a fixed length is examined (See Figure 3a). The inverted pendulum system comprises of a speaker system with adjustable frequency and amplitude, with an accelerometer being placed on the membrane to measure the acceleration of the pivot. Frequency at the signal generator is varied between 16 – 30 Hz (constrained by limitation of available equipment), and the amplitude is adjusted until the stable configuration is reached. The data can then be used to compare with the Landau and Lifshitz’s model to determine the match between the theoretical formulation and the experimental result, as well as observing any trend that deviates from the prediction. The behavior of two-pendulum system is also explored, where the stability for system of two pendulums of equal lengths are observed and analyzed (See Figure 3b).

2. Background and Theory

Having a mechanical model is a good first step toward exploring the behavior of such system. This rather unorthodox phenomenon is known as “Kapitza’s Pendulum”, and it is a classic example of a dynamical stabilization system [2][3][4]. A Kapitza’s pendulum can be modelled as an inverted pendulum under the effect of gravity, with the pivot being moved up and down periodically (see Figure 3a) [4]. The equation of motion can be derived using the Lagrangian method of energy, resulting in an expression relating the angle of pendulum with the periodic forcing movement. This equation represents the model used to describe the pendulum system, in

which through mathematical stability analysis, also shows that the pendulum can be stable vertically given the right input frequency and amplitude.

2.1 Equation of Motion

Deriving the equation of motion is an essential step in studying the behavior of this mechanical system. Due to the complexity of the motion, the Lagrangian method of energy is a convenient way to derive the equation of motion for the pendulum system [2] [8]. The system has one degree of freedom, and is described in term of the variable θ , representing the angle of the inverted pendulum with the vertical axis (See Figure 3a). The expression for kinetic and gravitational potential energy can be written with respect to this generalized coordinate system. The first step is to define the coordinate system and identify the location of center of mass of the pendulum with respect to the inertial frame. The x and y coordinates of the center of mass (COM) can be located as follow (*note that the expression y_{COM} contains two components, due to oscillating pivot*):

$$y_{drive} = A \cos(\omega t) \quad (1)$$

$$x_{COM} = \frac{l}{2} \sin \theta \quad (2)$$

$$y_{COM} = \frac{l}{2} \cos \theta + A \cos(\omega t) \quad (3)$$

Expression of kinetic (T) and potential energy (V) can then be written in term of the system's variables and constants. Since θ is used for the coordinate system, the energy equations should be written in term of θ . From there, the Lagrangian, L , can be calculated as a difference between kinetic and potential energy. The equation of motion can be derived by taking derivative operations of the Lagrangian with respect to the coordinate system, and the result is as shown [8]:

$$\ddot{\theta} + \frac{\frac{1}{2}}{\frac{1}{4} + \frac{I_{COM}}{ml^2}} \left(\frac{\omega^2 A \cos(\omega t)}{l} - \frac{g}{l} \right) \sin \theta = 0 \quad (4)$$

This is a form of non-linear ordinary differential equation, with a periodic forcing term. From this equation, the stability condition of the pendulum system can be investigated.

2.2 Stability Analysis

From the equation of motion, the stability of the pendulum can be studied. In Landau and Lifshitz's reference, it is shown the angular displacement can be described as a combination of a "main" trajectory $\theta_{main}(t)$ as well as small oscillation around the main trajectory $\Delta\theta_{vibrate}(t)$ [7]:

$$\theta(t) = \theta_{main}(t) + \Delta\theta_{vibrate}(t) = \theta_{main}(t) + C_1 \cos(\omega t) + C_2 \sin(\omega t) \quad (5)$$

It is shown that with an assumption about the input frequency ($\omega^2 \gg \frac{g}{l}$), as well as knowing that C_1 and C_2 are small and slowly varying, the following equation describing the “main” trajectory can be written as follow, using the derived equation of motion [8]:

$$\theta_{main}'' + \frac{1}{\frac{1}{4} + \frac{I_{COM}}{ml^2}} \left(\frac{\omega^2 A^2}{4l^2 \left(\frac{1}{4} + \frac{I_{COM}}{ml^2} \right)} - \frac{g}{l} \right) \theta_{main} = 0 \quad (6)$$

This is in the form of a second-order homogeneous differential equation. The solution to this equation is stable only when all of the coefficients are equal or greater than zero. As such, the coefficient in front of θ_{main} must satisfy the following condition for the pendulum to be stable [8]:

$$\begin{aligned} \frac{1}{\frac{1}{4} + \frac{I_{COM}}{ml^2}} \left(\frac{\omega^2 A^2}{4l^2 \left(\frac{1}{4} + \frac{I_{COM}}{ml^2} \right)} - \frac{g}{l} \right) &\geq 0 \\ \left(\frac{\omega^2 A^2}{4l^2 \left(\frac{1}{4} + \frac{I_{COM}}{ml^2} \right)} - \frac{g}{l} \right) &\geq 0 \\ \omega^2 A^2 &\geq 4gl \left(\frac{1}{4} + \frac{I_{COM}}{ml^2} \right) \end{aligned} \quad (7)$$

This result is consistent with the equation given in Landau and Lifshitz’s book, in which the equation for an inverted pendulum with a point mass is derived [7]. For a rod, $I_{COM} = \frac{1}{12}ml^2$, and the equation can be written as:

$$(A\omega)^2 \geq \frac{4}{3}gl \quad (8)$$

This is the key equation coming from the model that decides the stability of the system when the pendulum is in the inverted position. From the equation, it can be seen that if the combination of amplitude A and frequency ω is sufficiently large, the pendulum will remain stable in the inverted configuration.

3. Experimental Design

The experimental design included data acquisition and data analysis process. The data was collected and analyzed for a range of frequency between 16 and 30 Hz using a speaker and pendulum setup, and follow a specific sequence of analysis. There are two main components, including the data acquisition process and data analysis process. (See Figure 5 below)

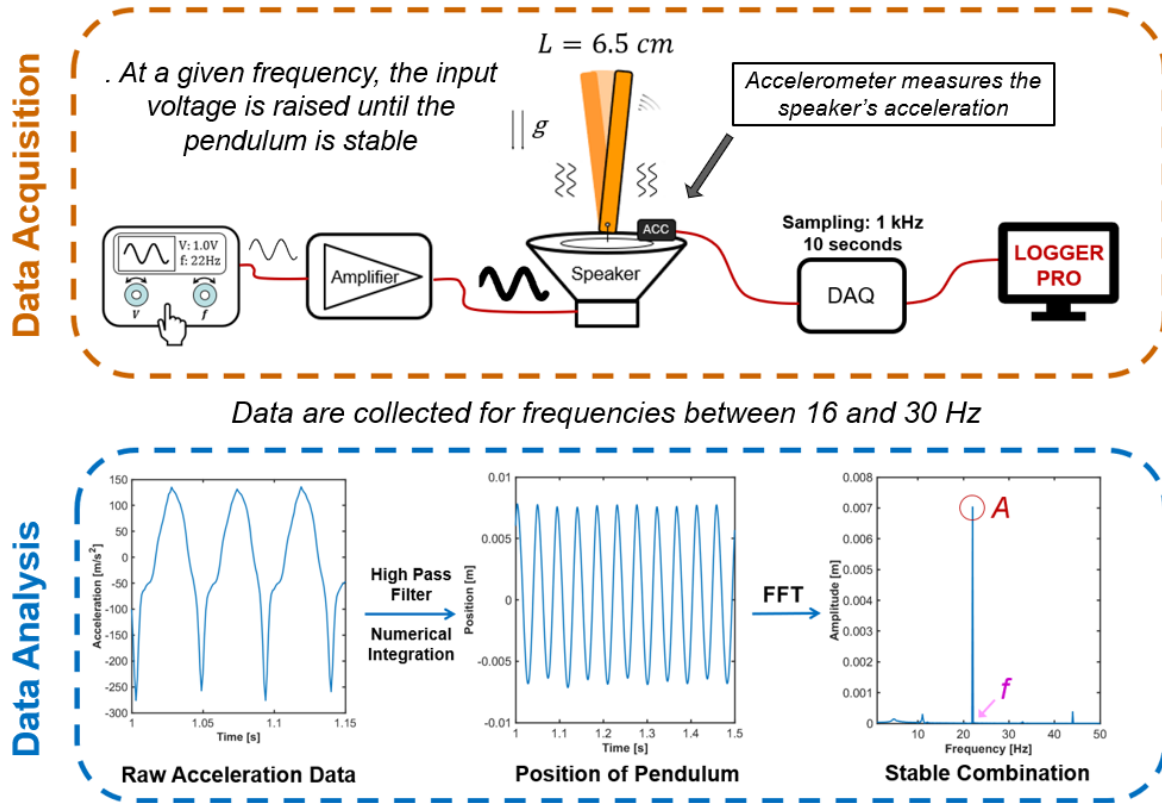


Figure 5: Experimental setup of the pendulum system. The process was divided into data acquisition and data analysis process. The process was repeated for each collection, and the stability result was plotted to compare with the empirical model.

The data acquisition process involves a custom speaker setup that is used to provide the oscillatory motion to the pendulum. Signal generator provides an adjustable sinusoidal signal to the amplifier, which is then used to drive the speaker system. An accelerometer is attached on the speaker membrane in order to measure the vertical acceleration of the speaker (which is also the acceleration of the pivot point of the pendulum). When the pendulum reaches the desired stable configuration, the accelerometer data collection is initiated. After the data is collected, the raw acceleration data is processed with MATLAB, where the acceleration data is filtered and numerically integrated. Fast Fourier Transform (FFT) algorithm is then used to obtain the pair of frequency and amplitude that make the pendulum stable. After the stability condition is collected for each of the attempt, the empirical model is obtained through curve fitting in comparison to the theory.

3.1 Definition of “Stability”

Before taking any measurement, defining the clear criteria for the stability of the pendulum is necessary, as it can be rather ambiguous as to what behavior is considered “stable”. For this pendulum setup, two different types are being recorded, naming “Global Stability” and “Local Stability”. The pendulum is globally stable when it’s able to reach the equilibrium from the laid-down position, without any external assistance. The pendulum is also able to eventually recover to the (inverted) stable point under any magnitude of disturbance impulse. On the other hand, the pendulum is locally stable if it can stay upright after being brought to the inverted position (but cannot get there itself from the dead hang position). The pendulum is *not* able to recover after the presence of significant disturbance.

3.2 Experimental Setup

The setup had two main parts, the actuator and the pendulum. A speaker setup was used to provide necessary actuation, and the pendulum was mounted to the speaker by a screw eye provided with the speaker. Vibration could then be applied to observe the behavior of the pendulum system.

3.2.1 Speaker System

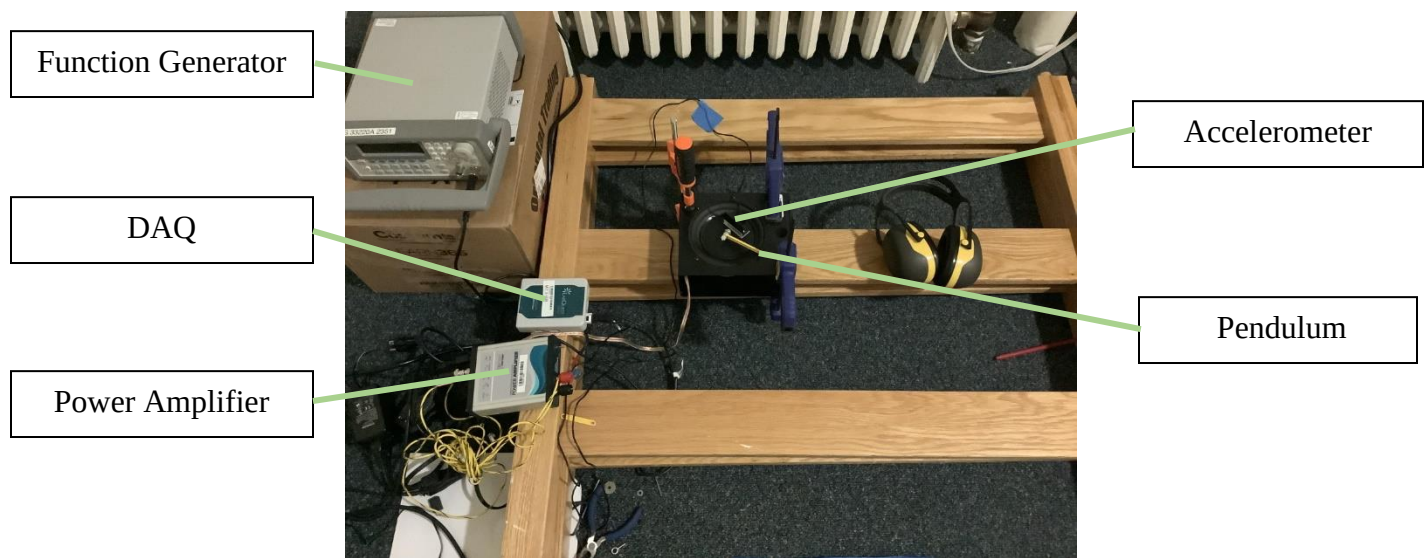


Figure 6: Speaker setup being assembled in dormitory room. Clamps are used to hold the speaker in place, with the speaker having the accelerometer and pendulum attached. The function generator on the top left corner is used to adjust input frequency and amplitude.

A speaker setup was created to provide necessary motion needed to vibrate the pendulum at a range of frequency (See Figure 5 and 6). This included the Vernier speaker (PAAS), the function generator FG 33220 (FG) as well as Vernier power amplification (POWER AMP). An accelerometer (ACC-25G) was also mounted on the speaker in order to measure the frequency and amplitude of the acceleration.

The power amplifier (POWER AMP) received an adjustable signal coming from the function generator (FG), and amplified this signal to drives the speaker (PAAS). The pendulum

stack-up was attached to the screw hole at the center of the speaker. The accelerometer (ACC) was mounted onto the surface of the speaker using an M3 double-sided tape. They were oriented in a way that allows for vertical acceleration measurement, while also not obstructing the pendulum's movement. ACC was then connected to the Data Acquisition Unit (LG - MINI), which in turns connect to the computer for data collection. The speaker was clamped into a disassembled bed frame, fixing the speaker in place while vibration is being applied to the pendulum.

There are a couple limitations on the system setup that affect the ability to measure data. The speaker setup is not able to provide enough power to sufficiently accelerate the pendulum system at higher frequency, resulting in a limited range of available frequency for testing. The accelerometer sensor on top of the speaker also adds mass on top of the speaker and reduces its ability to actuate the pendulum. Finally, the speaker's output for some frequencies contained higher harmonics of the desired frequency, which may have affected the position of pendulum in time in a way not accounted for by the theory.

3.2.2 Pendulum

The pendulums were attached to the speaker using a screw eye provided along with the speaker, and driven by the aforementioned speaker setup.



Figure 7: The stack-up of a single inverted pendulum, including hook, plastic rod, washer and M4 nylon screw and nuts. The pendulum has one degree of freedom, and is allowed to rotate freely around its pivot point. It is designed to be lightweight so the speaker can actuate with sufficient amplitude.



Figure 8: The stack-up of a multiple pendulum system. It had similar construction to the single pendulum counterpart, with an additional linkage attached.

The pendulum system, at a simple level, was a plastic rod that only has one degree of freedom (rotation around a pivot point). The speaker (PAAS) had a screw eye attachment that can be used to. Setting up a proper pendulum was crucial to the speaker's ability to provide sufficient vibration magnitude. The pendulum stack-up was designed to be as lightweight as possible to maximize the speaker's ability to accelerate the pendulum. For this setup, a combination of hardware was used to create the stack-up as shown in Figure 7: The plastic rod, found from a structure toy set, is used as a pendulum due to its light weight, as well as featuring a mounting hole. It had a length of

6.5 cm (center to center). M4 Nylon screw and nuts is used to hold the pendulum together. Nylon hardware were chosen as they are lighter than their metal counterparts, and there is no structural concern as all components are not bearing significant load.

The double pendulum setup was also implemented in a similar fashion, where an extra pendulum being connected to the tip of the first pendulum and was allowed to freely rotate. The two pendulums were at equal length of 3 cm, and they were attached in similar fashion to the single pendulum case, as shown in Figure 8.

3.3 Data Acquisition Method & Analysis

3.3.1 Data Acquisition Procedure

The acceleration was measured for the following set of frequencies between 16 to 30 Hz (See Figure 5). The acceleration data was collected 10 times for each frequency (5 trials for global stability, and 5 trials for local stability). This data was subsequently used to extract frequency and amplitude of the vibration. With the equipment being setup properly (see **3.1 Equipment Setup**), the collection process was started with setting up the computer software (Logger Pro) for data acquisition. The collection settings were set to 1000 samples per second, with 10 second duration. The function generator was set to a desired frequency and turned on at very low voltage output. The voltage to the power amplifier was then steadily increased until the desired state is achieved. For the global stability behavior, this output voltage was increased until the pendulum stays upright by itself. For the local stability behavior, the pendulum was raised to the upright position, and the voltage was raised until when the pendulum was released, it did not fall down. The acceleration data was then recorded into the computer, and the function generator is turned off to prepare for a new attempt.

3.3.2 Analyzing Collected Acceleration Data

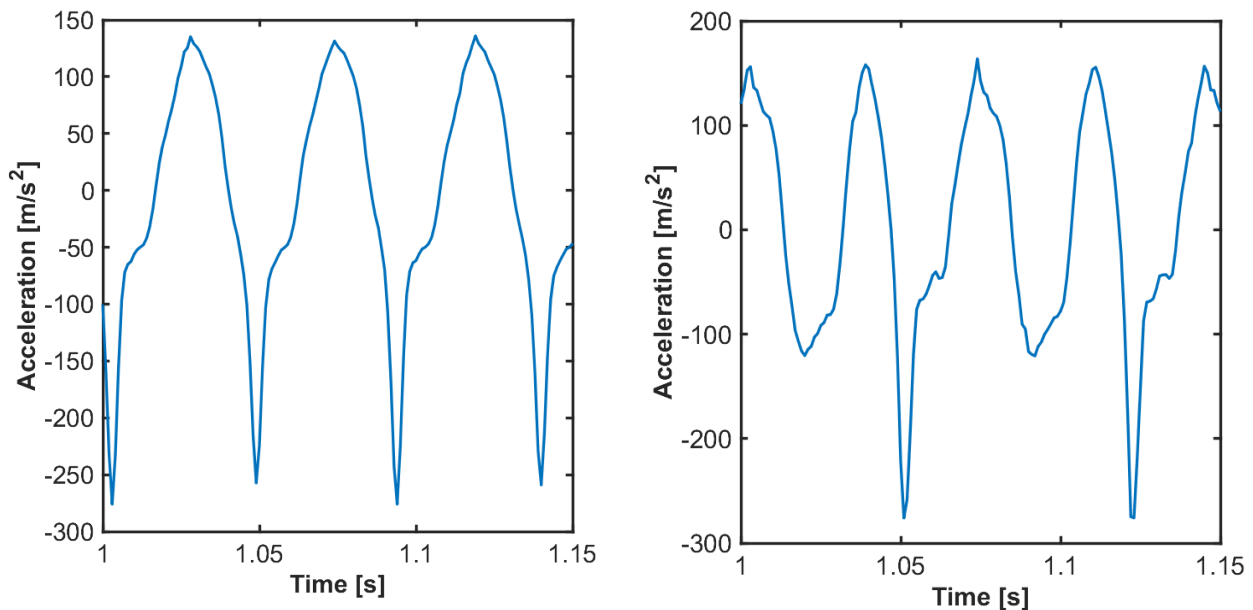


Figure 9: Representation of observed trend being seen from acceleration data (which is the vertical acceleration of the pivot). The left figure is seen for most acceleration behavior. The right figure shows acceleration data which has alternating downward peak. This has an implication on the stability analysis at that operating condition (see Figure 10).

The acceleration data was obtained for each of the result, and ready to be processed to obtain position measurement. For a strictly sinusoidal waveform (which is used by the empirical model), the position could be determined by multiplying the acceleration function by a factor of $\frac{1}{\omega^2}$. Since the resultant waveform was not a pure sinusoidal (see Figure 9), numerical integration is necessary to obtain the position function. The acceleration data was numerically integrated, in combination with high pass filter at cutoff frequency 5 Hz in order to obtain the position and remove any drift presented in the data. A custom fit (inverse function, $\frac{a}{x}$, where a is a constant factor) was used for curve-fitting purpose to compare with the empirical model. Each of the stability definition was then compared to the theoretical prediction to observe the relative stability.

3.3.3 Procedure for Two Pendulum System

The process for two pendulum system was mostly identical to the single pendulum, with an exception that only local stability behavior was measured and recorded. None of the recorded acceleration data for two pendulum system contain multiple harmonics, and thus are all suitable for analysis.

4. Results and Discussion

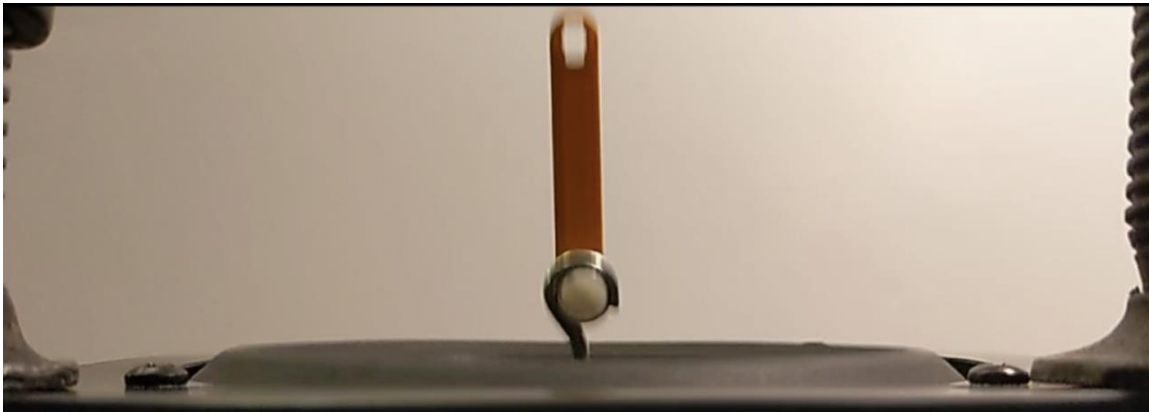


Figure 10: A snapshot of the pendulum system being stable in the inverted configuration. The pendulum manages to stay upright when sufficient vibration from the speaker is applied to the pivot.

4.1 Position Measurement

The position measurement behavior obtained from acceleration data is analyzed, which shows two observable trends, which have an implication on whether the data can be used. The first behavior features a relatively sinusoidal waveform with a single harmonic (see Figure 11). This behavior is appropriate for comparison with the theory, as the formulation only involves one specific frequency in the formula. This waveform is observed for most frequencies, except at 25 Hz. This characteristic can be seen with their corresponding FFT graph, where there's one apparent, strong peak at the input frequency, so these data can be used to compare with empirical model. The second behavior shows a waveform that is not sinusoidal, but instead containing more than 1 harmonics (see Figure 12). The FFT graph shows two significant harmonic (the first

harmonic, as well as a sub-harmonic) This waveform is observed for the trials at 25 Hz. This behavior can be attributed to the speaker's internal dynamics as the theory only predicts for the behavior of the pendulum under effect of one single frequency, data exerting this behavior are not valid to be used for curve-fitting purpose due to their multiple harmonics being presented. Thus, the data at 25 Hz is excluded from the curve fit analysis.

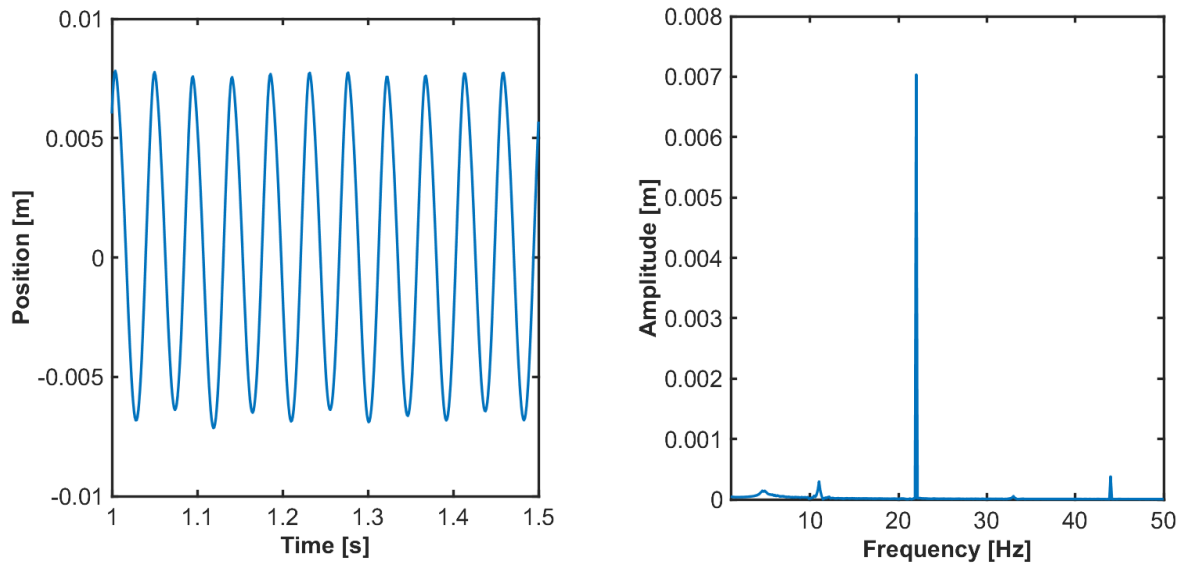


Figure 11: Position measurement shows a single harmonic, sinusoidal behavior over time. The amplitude and frequency of the signal is recorded in the frequency domain graph featured on the right

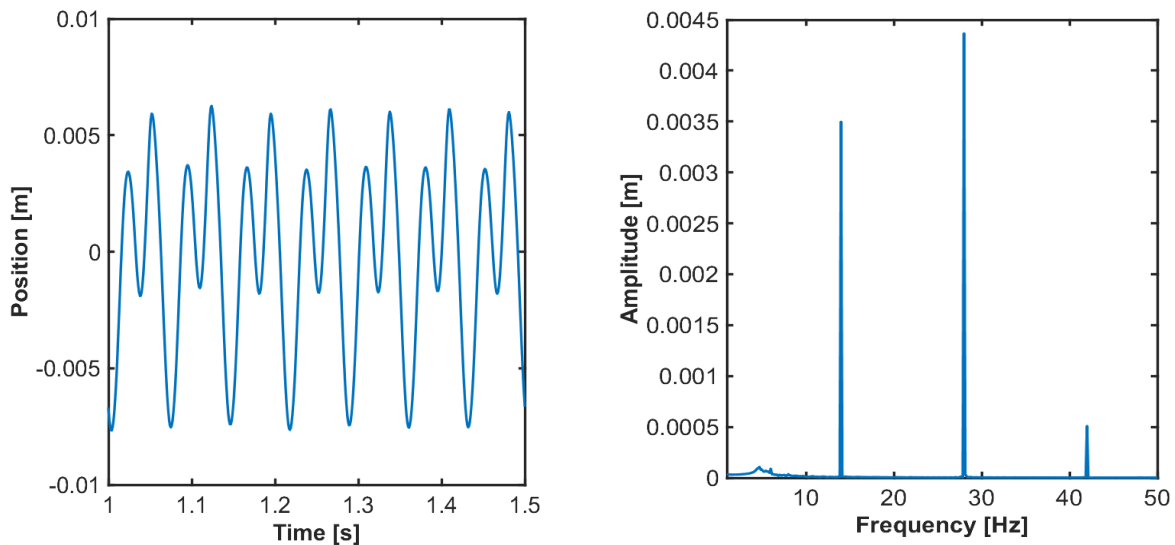


Figure 12: Position measurement shows multiple harmonics being observed in the signal. The data associated with this waveform is excluded from further analysis, as the theory does not predict the behavior of pendulum under multiple harmonic excitation (this can be a basis for potential experiments)

4.2 Single Pendulum Result

The combination of amplitude and frequency that stabilize the pendulum is obtained from the FFT graph, for each of the collection (See Figure 13). Inverse fit is then used to create the trendline for the experimental result.

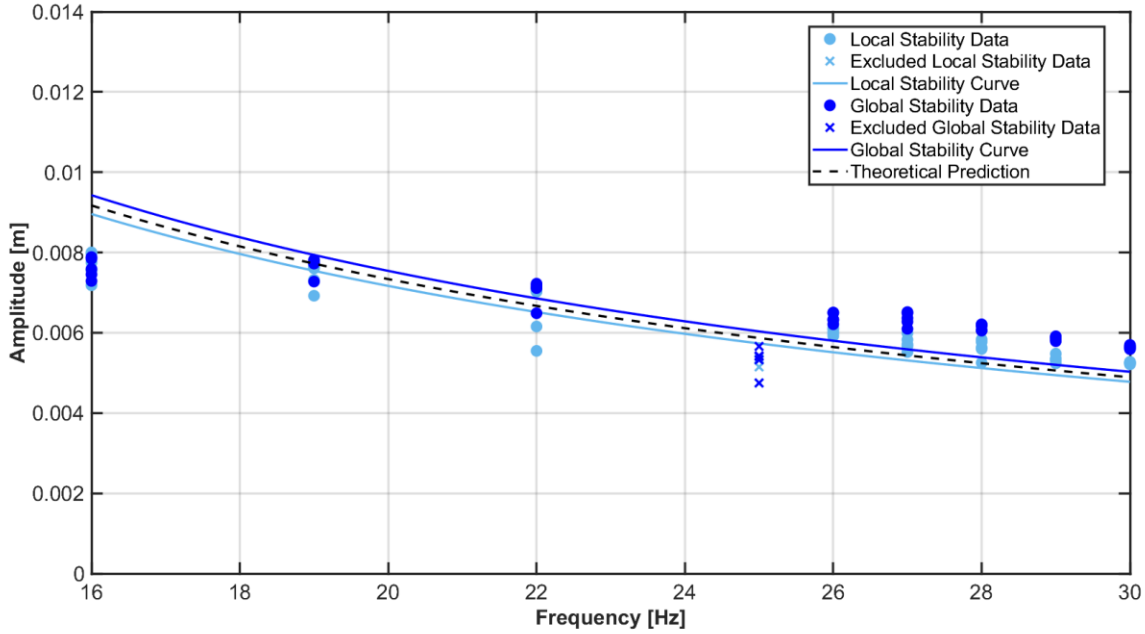


Figure 13: Graph showing experimental data and curve fit in comparison to the theoretical prediction. Inverse fit $\frac{a}{x}$ is done on all of the three stability conditions. The proportional constant for the: $a = 0.1420 \pm 0.0045 [m Hz]$; Strong Stability: $a = 0.1491 \pm 0.0060 [m Hz]$. The expected value of a is 0.1468, which is inside of the 95% confidence interval for all of the fit trial. Data at 25 Hz are excluded, as they contain more than one harmonic.

For the pendulum length of $L = 6.5\text{cm}$ and gravitational field strength of $g = 9.81 \frac{m}{s^2}$, the following function describes the theoretical curve constant using equation (10):

$$a_{theory} = \frac{\sqrt{\frac{4}{3}gl}}{2\pi} = 0.1468 [m Hz]$$

The measured data for the frequency range between 16 to 30 Hz (with the exception of 25Hz) are used for curve-fitting purpose, and plotted on the same graph (see Figure 11). The constant factor for each of them is as shown:

$$\text{Local Stability: } a_{local} = 0.1420 \pm 0.0045 [m Hz]$$

$$\text{Global Stability: } a_{global} = 0.1491 \pm 0.0060 [m Hz]$$

The overall shapes of all stability conditions show consistency with the predicted theory, and all of the fit parameters are statistically significant, where the uncertainty value is less than

magnitude of the parameter). The included uncertainty represents the 95% confident interval computed from the equation fit. For local stability, the 95% confidence interval of constant value does not contain the empirical model. For the global stability measurement, the 95% confident interval of proportional constant overlap with the predicted value of $a_{theory} = 0.1468$. The local stability constant value is 3% smaller than the empirical model, and the global stability constant value is 2% larger than empirical model. The theoretical prediction lies in between the two stability definitions ($a_{local} < a_{theory} < a_{global}$).

Thus, the empirical model provides a good model to predict the stability of the pendulum, showing a consistent trend with the empirical model and provide a good prediction to the stability of the pendulum system. As such, the linear model provides a simple way to quickly approximate and predict the stability behavior of such system.

4.3 Addendum – Exploration of Double Pendulum System

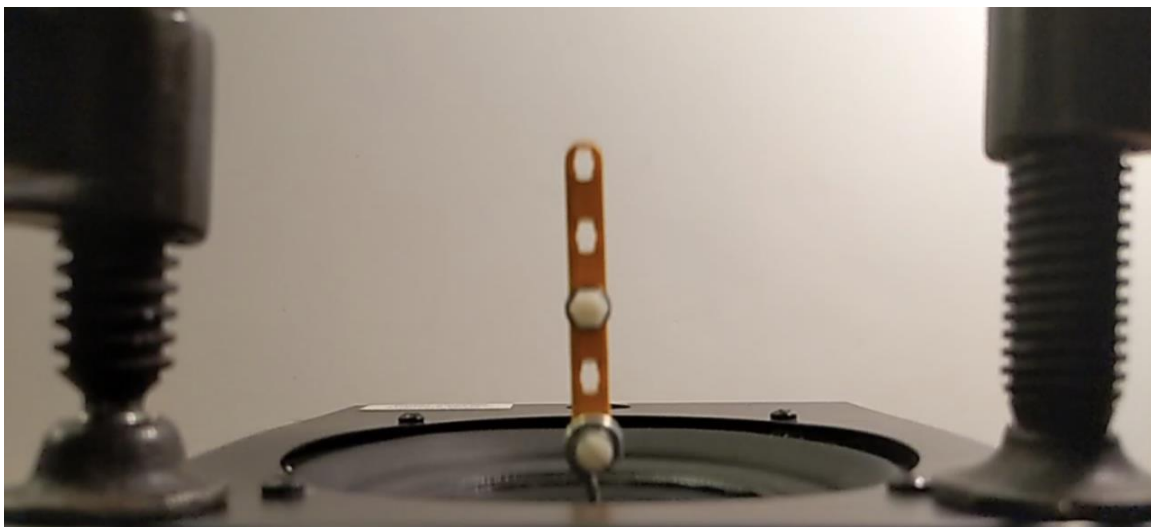


Figure 14: A snapshot of the double pendulum system being stable from the experiment. Both pendulums stay upright when sufficient vibration is applied

The stability of the two pendulums system is measured to explore the stability behavior of such system. The main observable stability behavior for the two pendulums system is the local stability condition, with the second pendulum needing assistance to reach equilibrium point. From the collected data, the stability of two pendulum system shows a sudden change in required magnitude after 28 Hz, behaving like a discontinuity (See Figure 15 below). The data shows an inverse trend at frequency above 28 Hz, similar to the trend for 1 pendulum system.

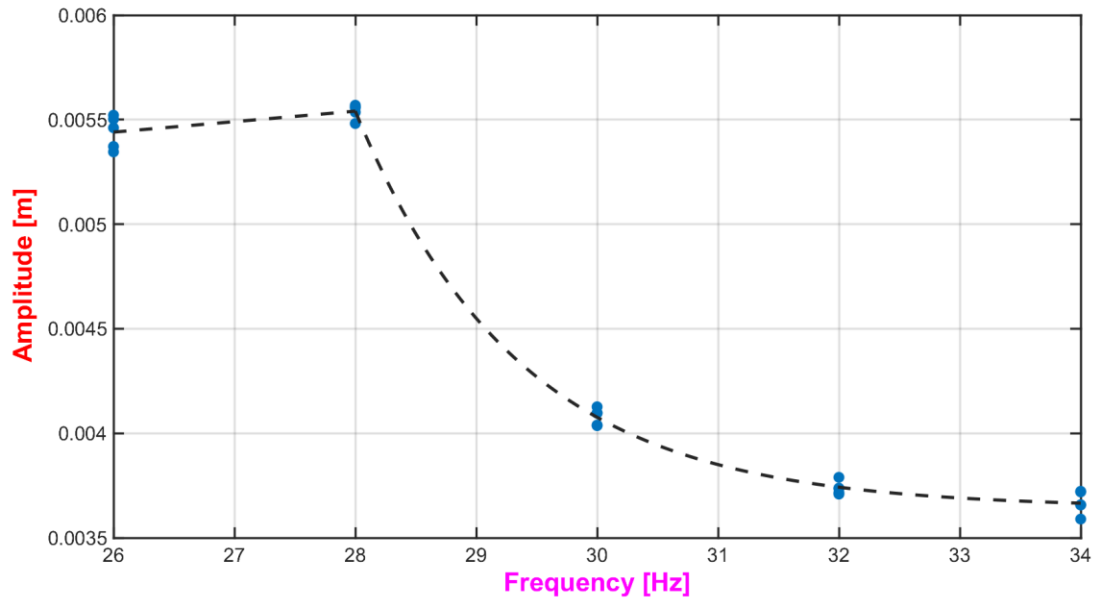


Figure 15: Two pendulums system stability result for $L = 3\text{cm}$. The pendulum system mostly observes local stability behavior, with the 2nd pendulum requiring assistance to reach stability condition. There is a sudden change in the data trend at 28 Hz. Trendline is showed in dashed line.

5. Conclusions

A pendulum system is created to compare with a current empirical model which predicts the stability of a pendulum under the effect of vibration. The result shows an agreement with empirical model in term of trend and magnitude for constant of fit for the pendulum length of $L = 6.5\text{cm}$. For local stability, the constant of proportionality for inverse fit is $a_{local} = 0.1420 \pm 0.0045 [m\text{ Hz}]$. For global stability, the constant of proportionality is $a_{global} = 0.1491 \pm 0.0060 [m\text{ Hz}]$. The empirical prediction of $a_{theory} = 0.1468 [m\text{ Hz}]$, which lies in between the two stability conditions. The theoretical constant also lies within the 95% confidence interval of global stability constant.

The difference between theoretical and experimental result is small, with local stability constant value being 3% smaller than the empirical model, and the global stability constant value being 2% larger. Thus, the model is able to predict the stability of the pendulum with high accuracy, with the theory lying in between local stability and global stability condition. More data at higher frequencies would be desirable in order to compare the theory prediction at these range of frequencies, in order to make a more robust statement on the accuracy of presented linear theory.

The stability experiment for two pendulum system was also conducted. The pendulum system manages to be stable, a discontinuous trend at 28 Hz, with an inverse function trend beyond 28Hz. More experiment can be conducted to explore the stability criterion of such fascinating system and understand how these multiple degree of freedom systems can be stable given an appropriate vibration excitation.

Acknowledgments

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References

- [1] Brad Bachu, “8.223 Project: Kapitza Pendulum”, *YouTube*, 29-Jan.-2015. [Online]. Available <https://www.youtube.com/watch?v=jS-rzZJovm4>. [Accessed: 06-Dec.-2021].
- [2] Phelps, F. M., and Hunter, J. H., 1965, “An Analytical Solution of the Inverted Pendulum,” *American Journal of Physics*, **33**(4), pp. 285–295.
- [3] Butikov, E. I., “Kapitza’s Pendulum: A Physically Transparent Simple Treatment,” p. 20.
- [4] Marin Bukov, Luca D’Alessio, and Anatoli Polkovnikov. Universal high-frequency behavior of periodically driven systems: from dynamical stabilization to floquet engineering. *Advances in Physics*, 64(2):139–226, 2015.
- [5] Apffel, B., Novkoski, F., Eddi, A., and Fort, E., 2020, “Floating under a Levitating Liquid,” *Nature*, **585**(7823), pp. 48–52.
- [6] Emmanuel Fort, “Creation of a levitating liquid layers”, *YouTube*, 03-Oct.-2020. [Online]. Available: <https://www.youtube.com/watch?v=3eGt3fClJAM>. [Accessed: 03-Oct.-2021].
- [7] Jones, P. H., Smart, T. J., Richards, C. J., and Cubero, D., 2016, “Optical Kapitza Pendulum,” *Optical Trapping and Optical Micromanipulation XIII*, SPIE, p. 992202.
- [8] L. D. Landau and E. M. Lifschitz, *Mechanics*, Pergamon, New York (1976), pp. 93 – 95.
- [9] L. D. Landau and E. M. Lifschitz, *Mechanics*, Pergamon, New York (1976), pp. 11 – 13.
- [10] Lester, C., 2017. [online] [Hep.phy.cam.ac.uk](https://www.hep.phy.cam.ac.uk). Available at: <https://www.hep.phy.cam.ac.uk/~lester/teaching/PartIIIProjects/2017-PeterDolan-InvertedPendulum.pdf> [Accessed 2 October 2021].